1 Gain law

($e_\theta = \theta - \hat{\theta}$; coefficients at the estimate)

$$\bar{a}(\bar{w}, b) = 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)}$$
$$\bar{a}' := \frac{\partial \bar{a}}{\partial \bar{w}} = b\big(1 - \bar{a}\big)^2$$
$$\bar{a}'' := \frac{\partial \bar{a}'}{\partial \bar{w}} = -2\,b^2\big(1 - \bar{a}\big)^3$$
$$A_a := \frac{\partial \bar{a}'}{\partial \bar{a}} = -2\,b\big(1 - \bar{a}\big)$$
$$\Big(\frac{\partial \bar{a}'}{\partial b}\Big)_{\bar{a}} = \big(1 - \bar{a}\big)^2$$
$$\Big(\frac{\partial \bar{a}}{\partial b}\Big)_{\bar{w},\,\bar{w}_s} = \frac{1 - \bar{a}}{b}$$

2 State

($\alpha = 1 - \lambda_c$)

$$\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b,\ x_6,\ x_7,\ m_1,\ m_2\,\big]^\top$$
$$\mathbf{e} = \big[\,e_1,\ e_2,\ e_3,\ e_4,\ e_b,\ e_6,\ e_7,\ e_8,\ e_9\,\big]^\top, \qquad e_j = e_{\delta\bar{w}_j}\ (j \le 3), \quad e_4 = e_{\bar{a}_w}$$

3 Substitution

$$e_{\bar{a}_w}[k] = \psi[k] := -\hat{b}\big(1 - \hat{\bar{a}}_w\big)^2 e_3[k] + \frac{1 - \hat{\bar{a}}_w}{\hat{b}}\, e_b[k]$$

4 Error dynamics

$$F_{12} = F_{23} = F_{55} = F_{66} = F_{77} = F_{98} = 1$$

Original

$$F_{33} = \lambda_c, \qquad F_{34} = -F_{dw}, \qquad F_{35} = 0, \qquad F_{38} = F_{39} = -\alpha$$
$$F_{43} = (1-\lambda_c)\,\bar{a}' + J_{d3}M$$
$$F_{44} = 1 + \bar{a}' F_{dw} + A_a M$$
$$F_{45} = \big(1-\hat{\bar{a}}_w\big)^2 M$$
$$F_{48} = F_{49} = \bar{a}'\alpha$$

Green

$$F_{33} = \lambda_c + F_{dw}\,\bar{a}', \qquad F_{34} = 0, \qquad F_{35} = -F_{dw}\,\frac{1-\hat{\bar{a}}_w}{\hat{b}}, \qquad F_{38} = F_{39} = -\alpha$$
$$F_{43} = (1-\lambda_c)\,\bar{a}' + J_{d3}M - F_{dw}\,\bar{a}'^2$$
$$F_{44} = 1 + A_a M$$
$$F_{45} = \big(1-\hat{\bar{a}}_w\big)^2 M + F_{dw}\big(1-\hat{\bar{a}}_w\big)^3$$
$$F_{48} = F_{49} = \bar{a}'\alpha$$

Both

$$\mathbf{g}_n = \big[\,0,\ 0,\ -(1-\lambda_c),\ \bar{a}'(1-\lambda_c),\ 0,\ 0,\ 0,\ 0,\ 0\,\big]^\top$$
$$F_{:,1} \leftarrow F_{:,1} - \mathbf{g}_n$$
$$\mathbf{g}_T = \big[\,0,\ 0,\ -1,\ \bar{a}',\ 0,\ 0,\ 0,\ 1,\ 0\,\big]^\top$$
$$Q = \kappa_T \hat{\bar{a}}_w\, \mathbf{g}_T \mathbf{g}_T^\top$$

5 Step error

$$u[k] = \mathbf{c}[k]^\top \mathbf{e}[k] + w_T[k], \qquad \mathrm{Var}\big(w_T[k]\big) = \kappa_T \hat{\bar{a}}_w[k]$$
$$\mathbf{c}_{\text{orig}} = \Big[-(1-\lambda_c),\ 0,\ (1-\lambda_c),\ F_{dw},\ 0,\ 0,\ 0,\ \alpha,\ \alpha\Big]^\top$$
$$\mathbf{c}_{\text{green}} = \Big[-(1-\lambda_c),\ 0,\ (1-\lambda_c) - F_{dw}\,\bar{a}',\ 0,\ F_{dw}\,\frac{1-\hat{\bar{a}}_w}{\hat{b}},\ 0,\ 0,\ \alpha,\ \alpha\Big]^\top$$
$$\mathrm{Cov}\big(e_j, u\big) = \big(P\,\mathbf{c}\big)_j, \qquad \mathrm{Var}(u) = \mathbf{c}^\top P\,\mathbf{c} + \kappa_T \hat{\bar{a}}_w$$

6 Second-order moments (green)

$$\operatorname{Cov}\big(e_3,u\big) = (1-\lambda_c)\big(P_{33}-P_{13}\big) + \alpha\big(P_{38}+P_{39}\big) + F_{dw}\Big(-\bar{a}'P_{33} + \frac{1-\hat{\bar{a}}_w}{\hat{b}}P_{35}\Big)$$

$$\operatorname{Cov}\big(e_b,u\big) = (1-\lambda_c)\big(P_{35}-P_{15}\big) + \alpha\big(P_{58}+P_{59}\big) + F_{dw}\Big(-\bar{a}'P_{35} + \frac{1-\hat{\bar{a}}_w}{\hat{b}}P_{55}\Big)$$

$$\operatorname{Cov}\big(\psi,u\big) = -\bar{a}'\operatorname{Cov}\big(e_3,u\big) + \frac{1-\hat{\bar{a}}_w}{\hat{b}}\operatorname{Cov}\big(e_b,u\big)$$

$$\begin{aligned}\operatorname{Var}(u) = {} & (1-\lambda_c)^2\big(P_{33}-2P_{13}+P_{11}\big) + \alpha^2\big(P_{88}+P_{99}+2P_{89}\big)\\ & + 2(1-\lambda_c)\,\alpha\big(P_{38}+P_{39}-P_{18}-P_{19}\big)\\ & + F_{dw}^2\Big(\bar{a}'^2P_{33} - 2\,\bar{a}'\,\frac{1-\hat{\bar{a}}_w}{\hat{b}}P_{35} + \Big(\frac{1-\hat{\bar{a}}_w}{\hat{b}}\Big)^2P_{55}\Big)\\ & - 2F_{dw}\,\bar{a}'\big((1-\lambda_c)(P_{33}-P_{13}) + \alpha(P_{38}+P_{39})\big)\\ & + 2F_{dw}\,\frac{1-\hat{\bar{a}}_w}{\hat{b}}\big((1-\lambda_c)(P_{35}-P_{15}) + \alpha(P_{58}+P_{59})\big)\\ & + \kappa_T\hat{\bar{a}}_w\end{aligned}$$

7 Predict of the gain slot

$$\Delta\hat{\bar{w}}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] + \alpha\big(\hat{m}_1[k]+\hat{m}_2[k]\big) + (1-\lambda_c)\big(y_1[k]-\delta\hat{\bar{w}}_1[k]\big)$$

Original

$$\hat{\bar{a}}_w[k+1] = 1 - \frac{1}{1/\big(1-\hat{\bar{a}}_w[k]\big)+\hat{b}[k]\,\Delta\hat{\bar{w}}[k]} + A_a\operatorname{Cov}\big(e_{\bar{a}_w},u\big) + \big(1-\hat{\bar{a}}_w\big)^2\operatorname{Cov}\big(e_b,u\big) + \tfrac{1}{2}\,\bar{a}''\operatorname{Var}(u)$$

Green

$$\begin{aligned}\hat{\bar{a}}_w[k+1] &= 1 - \frac{1}{1/\big(1-\hat{\bar{a}}_w[k]\big)+\hat{b}[k]\,\Delta\hat{\bar{w}}[k]} + A_a\operatorname{Cov}\big(\psi,u\big) + \big(1-\hat{\bar{a}}_w\big)^2\operatorname{Cov}\big(e_b,u\big) + \tfrac{1}{2}\,\bar{a}''\operatorname{Var}(u)\\ &= 1 - \frac{1}{1/\big(1-\hat{\bar{a}}_w[k]\big)+\hat{b}[k]\,\Delta\hat{\bar{w}}[k]} - \bar{a}''\operatorname{Cov}\big(e_3,u\big) - \big(1-\hat{\bar{a}}_w\big)^2\operatorname{Cov}\big(e_b,u\big) + \tfrac{1}{2}\,\bar{a}''\operatorname{Var}(u)\end{aligned}$$

8 Unchanged

$$H^{\text{green}} = H, \qquad Q^{\text{green}} = Q, \qquad R^{\text{green}} = R, \qquad P^{\text{green}}[0] = P[0]$$